

An SU(3) Dirac Quark Carrier Wavefunction with Embedded Color Structure

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Abstract

This paper develops a new SU(3) carrier wave for the quark sector in which strong/color structure is embedded directly into the phase content of the wavefunction rather than appended afterward as an external label. Beginning from the ordinary Dirac carrier $u(x)A(x)e^{i\theta(x)}$, one new perpendicular internal phase generator j is introduced and the quark carrier is taken in the lean form

$$\Psi_q(x) = u(x)A(x) \left(e^{i\theta(x)} + j \sin\left(\frac{\theta(x)}{4}\right) \right),$$

with algebra

$$i^2 = j^2 = -1, \quad ij = ji, \quad (ij)^2 = +1.$$

The construction is then tested step by step. The carrier is shown to possess 8π closure, to generate three emergent color classes as phase-separated sectors of the new j -component, and to produce anti-color by sign inversion. Mesons arise as pairwise neutralization of the new sector, while baryons arise as triplet closure of three distinct phase classes. From these classes an extracted color vector is obtained, and the standard 3×3 SU(3)-type generator basis is shown to act naturally on this wavefunction-derived color space. Finally, the carrier is tested under Dirac propagation and is shown to remain closed in the finite algebra $\text{span}\{1, i, j, ij\}$, with the mixed ij -sector generated dynamically rather than inserted by hand.

The result is a new internally viable SU(3)-type carrier for quarks. In this construction, meson neutralization, baryon triplet closure, extracted color representation, and Dirac-compatible propagation all emerge from one common phase-carrier logic. The proposal therefore preserves the successful Dirac shell while extending it into a deeper quark/color sector from which the familiar strong-interaction structure can be recovered in wavefunction form.

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1 Introduction

The Dirac wavefunction remains one of the great structural successes of modern physics. [1, 2, 3], In its ordinary form,

$$\psi_{\text{Dirac}}(x) = u(x) A(x) e^{i\theta(x)},$$

it provides a coherent relativistic carrier for fermionic propagation, spinorial structure, and the observable residue of charged matter. In this sense, Dirac is not merely a convenient formal shell. It is the phase-carrying object that made the lepton sector intelligible. The present paper begins from the view that this success should be preserved, not discarded. The question is whether the quark sector requires a deeper carrier built from the same spirit rather than from an appended color label alone.

In standard quark theory, the usual route is to retain the ordinary Dirac carrier and then attach color as an internal SU(3) index. This representation is mathematically effective, but it leaves open a deeper structural question. Is color fundamentally a label attached to an already complete carrier, or should it arise from the wavefunction itself in the same way that the original phase carrier gives rise to spinorial structure? This paper pursues the second possibility. Its central claim is that the quark sector requires a new carrier wave in which the strong/color structure is embedded directly into the phase content of the wavefunction. [4, 5, 6, 7, 8]

The guiding idea comes from a simple recursive phase rule. Each newly added internal phase layer must satisfy two conditions: it must be perpendicular to the previously active directions, and

it must propagate at half the phase rate of the previously added internal phase layer.¹ In the earlier SU(2)/SU(3) coherence picture, this recursive rule generated the closure ladder associated with successive internal symmetry structure. [9] The ordinary carrier already realizes the earlier closure step. The present work asks what happens when one more perpendicular internal phase generator is added at the next recursive level.

The answer proposed here is intentionally lean. A new imaginary generator j is introduced and the quark carrier is taken to be

$$\Psi_q(x) = u(x) A(x) \left(e^{i\theta(x)} + j \sin\left(\frac{\theta(x)}{4}\right) \right),$$

with algebra

$$i^2 = j^2 = -1, \quad ij = ji, \quad (ij)^2 = +1.$$

This construction is not introduced as an aesthetic embellishment. It is introduced because it can be tested against the minimal structural demands of the quark sector. A viable carrier must realize the next closure rung, produce three distinct phase classes without primitive RGB assignment, support pairwise neutralization and triplet closure, admit extraction of a genuine color-space basis, and remain compatible with Dirac propagation. These are strong enough requirements that a merely decorative extension would fail quickly.

What distinguishes the present approach is that the color matrix is not assumed at the beginning. Instead, the carrier first generates three equally separated phase classes from the new j -sector. These classes provide pairwise closure for mesons and collective triplet closure for baryons. Only after this wavefunction-level structure is established is the familiar 3×3 SU(3)-type matrix representation pulled out of the carrier. [10] In this way, the standard matrix language is preserved, but its role is changed: it becomes a representation of an already-real carrier structure rather than the primary source of that structure.

The final and decisive test is propagation. If the new carrier is only formal decoration, the Dirac equation will drive it outside its own algebra. [1, 3] If, however, the carrier remains closed under Dirac propagation and generates the mixed sector dynamically, then the proposal has passed the first serious viability test. That is the precise aim of this paper. It does not claim to solve all of QCD or all of hadronic structure. Rather, it establishes a new quark carrier, tests it against closure, meson and baryon formation, extracted SU(3)-type matrix action, and Dirac compatibility, and shows that the observable Dirac shell can be recovered as the residue of composite closure.

The result is a new internally viable SU(3) carrier wave for quarks. In this construction, meson neutralization, baryon triplet closure, extracted color representation, and Dirac-compatible propagation emerge from one common phase-carrier logic. Dirac is therefore not replaced. It is extended into a deeper quark/color sector from which the familiar strong-interaction structure can be recovered in wavefunction form.

2 Algebraic Setup and Carrier Definition

This section prepares the mathematical setup for the derivation that follows. Its role is not to perform the full argument, but to define the carrier, its algebra, and the minimal extracted objects

¹The half-rate propagation rule is not an arbitrary assumption. It follows from the coherence-locking mechanism developed in [9], where successive internal phase layers arise through recursive stability constraints. In that framework, each newly admitted phase degree of freedom must evolve at a reduced rate relative to the previously active layer in order to maintain global phase coherence and closure. The present construction adopts this rule as the next step in that hierarchy.

that Section 3 will test in detail. All milestone proofs—closure, meson neutralization, baryon triplet closure, matrix extraction, and Dirac propagation—are reserved for the next section.

2.1 The ordinary carrier as the starting point

The starting point is the ordinary complex phase carrier

$$A(x)e^{i\theta(x)},$$

which may be written in the familiar form

$$A(x)e^{i\theta(x)} = A(x) (\cos \theta(x) + i \sin \theta(x)).$$

[1, 2] In the present framework this object is taken as the base carrier from which the next internal phase layer will be constructed. The quantities appearing here are:

$A(x)$: real carrier amplitude,

$\theta(x)$: carrier phase variable,

i : ordinary imaginary phase generator.

The ordinary Dirac shell is then built in the standard way by placing this carrier inside a spinorial field,

$$\psi_{\text{Dirac}}(x) = u(x) A(x) e^{i\theta(x)},$$

where $u(x)$ is the Dirac spinor.

2.2 Recursive phase rule

The present construction adopts the same recursive phase rule used in the earlier coherence picture.

Each newly added internal phase layer must satisfy two conditions:

1. it must be perpendicular to the previously active directions;
2. it must propagate at half the phase rate of the previously added internal phase layer.

These two rules are the structural input of the construction. [9] They will be used in Section 3 to test whether the new carrier realizes the next closure rung and whether the resulting extracted phase classes support the quark color sector.

2.3 The new perpendicular phase generator

To move beyond the ordinary carrier, a new internal phase generator j is introduced. It is required to behave as a genuine phase generator rather than a symbolic label. The working algebra adopted in this paper is

$$\boxed{i^2 = j^2 = -1, \quad ij = ji, \quad (ij)^2 = +1.}$$

The associated finite carrier algebra is therefore

$$\boxed{\mathcal{A} = \text{span}\{1, i, j, ij\}.}$$

This is the minimal algebra needed for the proposed quark carrier. Its role is only to define the allowed sectors in which the wavefunction may propagate. The question of whether these sectors are sufficient is deferred to the tests in Section 3.

2.4 Definition of the new quark carrier

The new quark carrier is defined by

$$\Psi_q(x) = u(x) A(x) \left(e^{i\theta(x)} + j \sin\left(\frac{\theta(x)}{4}\right) \right).$$

The symbols are:

$u(x)$: Dirac spinor,

$A(x)$: real amplitude,

$\theta(x)$: carrier phase,

i : ordinary Dirac-like phase generator,

j : new perpendicular strong/color phase generator.

Expanding the carrier factor gives

$$\mathcal{E}(\theta) = e^{i\theta} + j \sin\left(\frac{\theta}{4}\right) = \cos \theta + i \sin \theta + j \sin\left(\frac{\theta}{4}\right).$$

Thus the input carrier visibly occupies three sectors,

$$1, \quad i, \quad j,$$

while the mixed sector ij is not inserted by hand at the carrier level. Whether that mixed sector arises dynamically is one of the central questions to be tested later.

2.5 Extracted phase classes

The new j -sector is not initially interpreted as a primitive RGB label. Instead, three phase classes are extracted from it by equally separated shifts:

$$c_0(\theta) = \sin\left(\frac{\theta}{4}\right), \quad c_1(\theta) = \sin\left(\frac{\theta}{4} + \frac{2\pi}{3}\right), \quad c_2(\theta) = \sin\left(\frac{\theta}{4} + \frac{4\pi}{3}\right).$$

Their opposite sectors are defined by sign inversion,

$$\bar{c}_k(\theta) = -c_k(\theta), \quad k = 0, 1, 2.$$

These objects are introduced here only as extracted phase classes. Their interpretation as color and anti-color sectors, together with their closure properties, will be established in Section 3.

For convenience, we also package them into the vector

$$\mathbf{c}(\theta) = \begin{pmatrix} c_0(\theta) \\ c_1(\theta) \\ c_2(\theta) \end{pmatrix} = \begin{pmatrix} \sin(\theta/4) \\ \sin(\theta/4 + 2\pi/3) \\ \sin(\theta/4 + 4\pi/3) \end{pmatrix}.$$

At this stage $\mathbf{c}(\theta)$ is simply the extracted phase-basis candidate on which the later matrix representation will act.

2.6 Purpose of the next section

With the carrier, algebra, and extracted phase classes now defined, the next section performs the full derivation. In particular, Section 3 will test:

- whether the new carrier realizes the next closure rung;
- whether the extracted phase classes support pairwise neutralization and triplet closure;
- whether the color-space basis yields a natural 3×3 matrix representation;
- whether the standard SU(3)-type generator basis acts consistently on that extracted space;
- and whether the full carrier remains closed under Dirac propagation.

Thus the present section supplies the setup only. The actual proof of viability belongs to the section that follows.

3 A New SU(3) Carrier Wave for the Quark Sector

The purpose of this section is to construct, test, and validate the new quark carrier introduced in Section 2. The role of the section is not merely to display a sequence of formal computations, but to show that the proposed carrier satisfies the minimum structural demands required of a viable quark-sector wavefunction.

Those demands are specific. First, the carrier must realize the next closure rung beyond the ordinary SU(2)-type carrier. Second, it must generate three distinct internal phase classes without assuming primitive RGB labels by hand. Third, those extracted classes must support the two observed hadronic closure modes: pairwise neutralization for mesons and collective triplet closure for baryons. Fourth, the extracted basis must support a natural 3×3 SU(3)-type matrix action. Finally, the carrier must remain compatible with the Dirac shell under propagation and remain closed in a finite algebra.

In other words, the section tests whether the proposal

$$\Psi_q(x) = u(x) A(x) \left(e^{i\theta(x)} + j \sin\left(\frac{\theta(x)}{4}\right) \right)$$

is merely suggestive or whether it is already strong enough to function as a genuine carrier candidate for the quark sector. Each milestone therefore has a precise role. The closure test establishes the new rung of the recursive phase ladder. The extracted classes establish the internal three-fold structure. The meson and baryon tests show that the same carrier supports both basic hadronic existence modes. The color-vector and matrix subsections show that the standard SU(3) representation can be recovered from the carrier rather than assumed at the start. The Dirac test then determines whether the whole construction survives propagation in a nontrivial but closed way.

3.1 Carrier legend and algebra

We begin by restating the carrier in the compact form used throughout this section:

$$\Psi_q(x) = u(x) A(x) \mathcal{E}(x), \quad \mathcal{E}(x) = e^{i\theta(x)} + j \sin\left(\frac{\theta(x)}{4}\right).$$

The symbols are:

- $u(x)$: Dirac spinor,
- $A(x)$: real amplitude,
- $\theta(x)$: carrier phase variable,
- i : original Dirac-like phase generator,
- j : new perpendicular color/strong phase generator.

The algebra used in the present construction is the minimal commuting extension

$$i^2 = j^2 = -1, \quad ij = ji, \quad (ij)^2 = +1.$$

This choice is crucial. It preserves i and j as the two pure phase generators while making the mixed sector ij structurally distinct from the pure sectors. The resulting carrier algebra is therefore

$$\mathcal{A} = \text{span}\{1, i, j, ij\}.$$

The carrier factor can also be written explicitly as

$$\mathcal{E}(\theta) = \cos \theta + i \sin \theta + j \sin\left(\frac{\theta}{4}\right).$$

Thus the carrier initially occupies three visible sectors,

$$1, \quad i, \quad j,$$

while the mixed sector ij is absent at the input stage and must emerge dynamically if the construction is viable. This is the first important structural feature of the ansatz: the mixed sector is not inserted by hand, so if it appears later it must do so as a consequence of propagation.

3.2 First milestone: closure test

The first question is whether the proposed carrier actually realizes the next rung of the recursive phase ladder. This test matters because the entire construction rests on the claim that the new internal phase layer should move the carrier beyond the earlier closure class. If this test fails, then the carrier is not genuinely new in the required sense.

We therefore test

$$\mathcal{E}(\theta) = e^{i\theta} + j \sin\left(\frac{\theta}{4}\right).$$

At $\theta + 4\pi$,

$$e^{i(\theta+4\pi)} = e^{i\theta},$$

but

$$\sin\left(\frac{\theta + 4\pi}{4}\right) = \sin\left(\frac{\theta}{4} + \pi\right) = -\sin\left(\frac{\theta}{4}\right).$$

Hence

$$\mathcal{E}(\theta + 4\pi) = e^{i\theta} - j \sin\left(\frac{\theta}{4}\right) \neq \mathcal{E}(\theta).$$

At $\theta + 8\pi$,

$$e^{i(\theta+8\pi)} = e^{i\theta},$$

and

$$\sin\left(\frac{\theta + 8\pi}{4}\right) = \sin\left(\frac{\theta}{4} + 2\pi\right) = \sin\left(\frac{\theta}{4}\right).$$

So

$$\mathcal{E}(\theta + 8\pi) = \mathcal{E}(\theta).$$

Therefore the new carrier is not 4π -closed, but is 8π -closed:

$$\boxed{\mathcal{E}(\theta + 8\pi) = \mathcal{E}(\theta).}$$

This is the first major pass of the construction. It shows that the new j -sector is not a decorative addition to the old carrier, but actually lifts it to the next closure class required by the recursive phase rule.

3.3 Second milestone: emergent color classes

Once the closure rung is established, the next question is whether the new sector contains a genuine three-fold internal structure. This is where the present route diverges most sharply from the standard label-first picture. We do not begin by assigning RGB as primitive basis labels. Instead, we ask whether three distinct phase classes can be extracted directly from the new j -sector.

Define

$$c_0(\theta) = \sin\left(\frac{\theta}{4}\right), \quad c_1(\theta) = \sin\left(\frac{\theta}{4} + \frac{2\pi}{3}\right), \quad c_2(\theta) = \sin\left(\frac{\theta}{4} + \frac{4\pi}{3}\right).$$

These are the three emergent color classes. Their opposite sectors are defined by sign inversion:

$$\bar{c}_k(\theta) = -c_k(\theta), \quad k = 0, 1, 2.$$

Thus color and anti-color are not separate primitive labels, but opposite sectors of one common extracted phase family. This is the first point at which the carrier begins to behave like a genuine quark-sector object rather than an ordinary Dirac carrier with extra notation attached.

3.4 Third milestone: meson closure

The next test asks whether the extracted classes support pairwise neutralization in the way required for mesonic structure. This matters because a viable carrier must not merely contain three classes; it must also allow the correct closure logic for quark–antiquark pairing.

The extracted color rule is

$$c_k + \bar{c}_k = 0.$$

Explicitly,

$$c_k(\theta) + \bar{c}_k(\theta) = c_k(\theta) - c_k(\theta) = 0.$$

Thus pairwise closure of the j -sector is automatic:

$$\boxed{c_k + \bar{c}_k = 0.}$$

At the carrier level, let

$$\Psi_q = uA \left(e^{i\theta} + j \sin\left(\frac{\theta}{4}\right) \right), \quad \Psi_{\bar{q}} = vA \left(e^{i\theta} - j \sin\left(\frac{\theta}{4}\right) \right).$$

Then

$$\Psi_q + \Psi_{\bar{q}} = uAe^{i\theta} + vAe^{i\theta} + uAj \sin\left(\frac{\theta}{4}\right) - vAj \sin\left(\frac{\theta}{4}\right).$$

At the color-neutralized level, the j -sector cancels and the residual observed structure is the ordinary Dirac-like part,

$$\Psi_M \sim Ae^{i\theta}(u + v).$$

Thus meson formation appears as partial closure of the new carrier: the strong/color sector neutralizes, while the observable Dirac-like residue survives. This is one of the central achievements of the construction.

3.5 Fourth milestone: baryon triplet closure

A viable quark carrier must also support baryons. This is the second major hadronic existence mode, and it is not pairwise. The present test therefore asks whether the same extracted phase structure that supports mesons also supports collective triplet closure.

Using the three extracted classes,

$$c_0(\theta) + c_1(\theta) + c_2(\theta) = \sin x + \sin\left(x + \frac{2\pi}{3}\right) + \sin\left(x + \frac{4\pi}{3}\right), \quad x = \frac{\theta}{4}.$$

Using the standard trigonometric identity,

$$\sin x + \sin\left(x + \frac{2\pi}{3}\right) + \sin\left(x + \frac{4\pi}{3}\right) = 0,$$

we obtain

$$\boxed{c_0(\theta) + c_1(\theta) + c_2(\theta) = 0.}$$

Thus baryon closure is automatic in the new carrier.

At the carrier level, define the three quark states

$$\Psi_k = u_k A \left(e^{i\theta} + j \sin\left(\frac{\theta + 8\pi k/3}{4}\right) \right), \quad k = 0, 1, 2.$$

Summing,

$$\sum_{k=0}^2 \Psi_k = Ae^{i\theta}(u_0 + u_1 + u_2) + jA \sum_{k=0}^2 u_k \sin\left(\frac{\theta + 8\pi k/3}{4}\right).$$

At the closure level, the j -sector sums to zero, leaving the residual observed part

$$\Psi_B \sim Ae^{i\theta}(u_0 + u_1 + u_2).$$

So the same carrier generates meson pair neutralization and baryon triplet closure. This is one of the strongest conceptual results of the section: the two basic hadronic existence modes arise from two distinct closure modes of one common quark carrier.

3.6 Fifth milestone: extracted color vector

Now that the three phase classes have been established and shown to support the two closure modes, they may be packaged into a genuine color vector:

$$\mathbf{c}(\theta) = \begin{pmatrix} c_0(\theta) \\ c_1(\theta) \\ c_2(\theta) \end{pmatrix} = \begin{pmatrix} \sin(\theta/4) \\ \sin(\theta/4 + 2\pi/3) \\ \sin(\theta/4 + 4\pi/3) \end{pmatrix}.$$

This is the first honest color basis extracted from the carrier rather than assumed beforehand. Its importance is that the carrier has now produced a linear color space on which matrix representation can legitimately act.

Because

$$c_0 + c_1 + c_2 = 0,$$

the vector satisfies

$$(1, 1, 1) \cdot \mathbf{c}(\theta) = 0.$$

Hence $\mathbf{c}(\theta)$ lies in the traceless color-neutral plane from the outset. This is an especially attractive result because it means the extracted basis already carries the essential closure property of the strong/color sector.

3.7 Sixth milestone: basic color matrices

Only after the carrier has produced the color basis does the matrix picture become legitimate. This is one of the defining differences of the present route. The purpose of this subsection is to show that the extracted basis naturally supports the basic triplet matrix action. [10]

We begin with the simplest cyclic permutation matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix},$$

which acts by

$$P\mathbf{c}(\theta) = \begin{pmatrix} c_1 \\ c_2 \\ c_0 \end{pmatrix}.$$

This generates the color cycle

$$c_0 \rightarrow c_1 \rightarrow c_2 \rightarrow c_0.$$

Next, the elementary swap matrix

$$S_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

acts as

$$S_{12}\mathbf{c}(\theta) = \begin{pmatrix} c_1 \\ c_0 \\ c_2 \end{pmatrix}.$$

A diagonal phase distinguisher is

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \quad \omega = e^{2\pi i/3}.$$

So the extracted basis already supports the basic clock-and-shift structure of the triplet sector. This is the point at which the standard $SU(3)$ -type matrix language becomes a natural representation of the carrier-derived color space.

3.8 Seventh milestone: full $SU(3)$ -type generator basis

Having established that the extracted basis supports the basic triplet action, we now recover the full 3×3 $SU(3)$ -type generator structure. [10, 8] This step matters because it shows that the carrier does not merely suggest a three-fold internal geometry. It supports the standard linear algebraic language ordinarily used to describe the color sector.

The elementary off-diagonal matrices are

$$\begin{aligned} E_{12} &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & E_{21} &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ E_{23} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, & E_{32} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \\ E_{13} &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & E_{31} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \end{aligned}$$

From these one forms the standard Hermitian mixing matrices:

$$\begin{aligned} \lambda_1 &= E_{12} + E_{21} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \lambda_2 &= -i(E_{12} - E_{21}) = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \lambda_4 &= E_{13} + E_{31} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \\ \lambda_5 &= -i(E_{13} - E_{31}) = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \\ \lambda_6 &= E_{23} + E_{32} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \end{aligned}$$

$$\lambda_7 = -i(E_{23} - E_{32}) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

The two traceless diagonal generators are

$$\lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Together these eight matrices act on the extracted carrier basis $\mathbf{c}(\theta)$, not on externally assigned RGB labels. This is one of the most important structural accomplishments of the section.

3.9 Eighth milestone: explicit matrix action on the extracted basis

To confirm that the extracted color space is not formal only, we now act with the generators directly on $\mathbf{c}(\theta)$. The point of this test is to show that the carrier-derived basis is a genuine basis for matrix action rather than a symbolic intermediary.

For example,

$$\begin{aligned} \lambda_1 \mathbf{c}(\theta) &= \begin{pmatrix} c_1 \\ c_0 \\ 0 \end{pmatrix}, & \lambda_2 \mathbf{c}(\theta) &= \begin{pmatrix} -i c_1 \\ i c_0 \\ 0 \end{pmatrix}, \\ \lambda_4 \mathbf{c}(\theta) &= \begin{pmatrix} c_2 \\ 0 \\ c_0 \end{pmatrix}, & \lambda_6 \mathbf{c}(\theta) &= \begin{pmatrix} 0 \\ c_2 \\ c_1 \end{pmatrix}. \end{aligned}$$

The diagonal generators act as

$$\begin{aligned} \lambda_3 \mathbf{c}(\theta) &= \begin{pmatrix} c_0 \\ -c_1 \\ 0 \end{pmatrix}, \\ \lambda_8 \mathbf{c}(\theta) &= \frac{1}{\sqrt{3}} \begin{pmatrix} c_0 \\ c_1 \\ -2c_2 \end{pmatrix}. \end{aligned}$$

Thus the full 3×3 matrix action is now grounded in an explicitly extracted wavefunction-level basis. Because these are the standard Gell-Mann forms, their commutator structure closes in the usual way. [10] For example,

$$[\lambda_1, \lambda_2] = 2i\lambda_3.$$

So once the extracted basis is admitted, the full SU(3)-type generator structure follows naturally.

3.10 Ninth milestone: Dirac propagation test

The preceding milestones established the new closure class, the three extracted phase sectors, the two hadronic closure modes, and the recovered SU(3)-type matrix action. The final major test is whether the carrier remains viable under Dirac propagation. This is decisive. If the new carrier is only a formal decoration, the Dirac operator will drive it outside its own algebra. [1, 3, 2] If it remains closed, then the construction has passed its strongest viability test.

Write

$$\Psi_q(x) = u(x) A(x) \mathcal{E}(x), \quad \mathcal{E}(x) = e^{i\theta(x)} + j \sin\left(\frac{\theta(x)}{4}\right).$$

We test

$$(i\gamma^\mu \partial_\mu - m)\Psi_q = 0.$$

Differentiate the carrier:

$$\partial_\mu \mathcal{E} = \partial_\mu (e^{i\theta}) + \partial_\mu \left(j \sin\left(\frac{\theta}{4}\right) \right).$$

Since j is constant,

$$\partial_\mu (e^{i\theta}) = i(\partial_\mu \theta) e^{i\theta},$$

and

$$\partial_\mu \left(j \sin\left(\frac{\theta}{4}\right) \right) = j \cos\left(\frac{\theta}{4}\right) \frac{1}{4} \partial_\mu \theta.$$

Hence

$$\partial_\mu \mathcal{E} = (\partial_\mu \theta) \left[ie^{i\theta} + \frac{j}{4} \cos\left(\frac{\theta}{4}\right) \right].$$

Expanding $ie^{i\theta}$,

$$ie^{i\theta} = i(\cos \theta + i \sin \theta) = i \cos \theta - \sin \theta,$$

so

$$\partial_\mu \mathcal{E} = (\partial_\mu \theta) \left[-\sin \theta + i \cos \theta + \frac{j}{4} \cos\left(\frac{\theta}{4}\right) \right].$$

Now

$$\partial_\mu \Psi_q = (\partial_\mu u) A \mathcal{E} + u(\partial_\mu A) \mathcal{E} + u A \partial_\mu \mathcal{E},$$

and therefore

$$(i\gamma^\mu \partial_\mu - m)\Psi_q = i\gamma^\mu (\partial_\mu u) A \mathcal{E} + i\gamma^\mu u(\partial_\mu A) \mathcal{E} + i\gamma^\mu u A \partial_\mu \mathcal{E} - m u A \mathcal{E}.$$

The new nontrivial piece is

$$i\gamma^\mu u A \partial_\mu \mathcal{E} = i\gamma^\mu u A (\partial_\mu \theta) \left[-\sin \theta + i \cos \theta + \frac{j}{4} \cos\left(\frac{\theta}{4}\right) \right].$$

Distributing the outer i ,

$$i(-\sin \theta) = -i \sin \theta, \quad i(i \cos \theta) = -\cos \theta, \quad i \left(\frac{j}{4} \cos\left(\frac{\theta}{4}\right) \right) = \frac{ij}{4} \cos\left(\frac{\theta}{4}\right).$$

So

$$i\gamma^\mu u A \partial_\mu \mathcal{E} = \gamma^\mu u A (\partial_\mu \theta) \left[-\cos \theta - i \sin \theta + \frac{ij}{4} \cos\left(\frac{\theta}{4}\right) \right].$$

This is the critical pass of the propagation test. The mixed ij -sector appears automatically from propagation itself.

3.11 Tenth milestone: sector decomposition

The Dirac equation acting on the new carrier remains inside the finite algebra

$$\mathcal{A} = \text{span}\{1, i, j, ij\}.$$

Accordingly, we may write

$$(i\gamma^\mu \partial_\mu - m)\Psi_q = \mathcal{S}_1 + i\mathcal{S}_i + j\mathcal{S}_j + ij\mathcal{S}_{ij}.$$

The scalar and i -sectors come from the original $e^{i\theta}$ carrier:

$$\mathcal{E} = e^{i\theta} + j \sin\left(\frac{\theta}{4}\right) = \cos\theta + i \sin\theta + j \sin\left(\frac{\theta}{4}\right).$$

The j -sector comes directly from the new color carrier:

$$j \sin\left(\frac{\theta}{4}\right).$$

The mixed ij -sector is not inserted by hand. It is generated automatically by propagation:

$$\mathcal{S}_{ij} = \frac{1}{4} \gamma^\mu u A(\partial_\mu \theta) \cos\left(\frac{\theta}{4}\right).$$

Thus the carrier algebra closes under Dirac propagation. No uncontrolled extra sectors appear. This is the decisive technical pass of the section.

3.12 Eleventh milestone: interpreted sector roles

The final step is to interpret the algebraic sectors that have now emerged. At this point the structure is no longer merely symbolic; each sector has acquired a clear role through the sequence of tests.

The four algebra sectors are interpreted as

$$\begin{aligned} 1, i &\Rightarrow \text{ordinary observable Dirac residue,} \\ j &\Rightarrow \text{new strong/color carrier sector,} \\ ij &\Rightarrow \text{mixed propagated quark sector.} \end{aligned}$$

This interpretation is exactly what the earlier milestones prepared. Mesons neutralize the j -sector and reduce to the $1/i$ residue. Baryons close the three j -phase classes and again reduce to the $1/i$ residue. Propagation then dynamically entangles the old and new sectors through ij . The resulting picture is therefore coherent from beginning to end: the carrier begins with $1, i, j$, and the full propagated quark structure closes naturally in the extended algebra $\{1, i, j, ij\}$.

3.13 Summary of the achieved result

We may now summarize the full achievement of the construction.

The new carrier

$$\Psi_q(x) = u(x) A(x) \left(e^{i\theta(x)} + j \sin\left(\frac{\theta(x)}{4}\right) \right)$$

with commuting algebra

$$i^2 = j^2 = -1, \quad ij = ji, \quad (ij)^2 = +1$$

successfully satisfies all of the intended tests.

First, it realizes the next closure rung:

$$\Psi_q(\theta + 8\pi) = \Psi_q(\theta).$$

Second, it generates three emergent color classes directly from the carrier:

$$c_0(\theta) = \sin(\theta/4), \quad c_1(\theta) = \sin(\theta/4 + 2\pi/3), \quad c_2(\theta) = \sin(\theta/4 + 4\pi/3).$$

Third, it gives pair neutralization:

$$c_k + \bar{c}_k = 0.$$

Fourth, it gives triplet closure:

$$c_0 + c_1 + c_2 = 0.$$

Fifth, it produces an extracted color basis

$$\mathbf{c}(\theta) = \begin{pmatrix} c_0(\theta) \\ c_1(\theta) \\ c_2(\theta) \end{pmatrix}, \quad (1, 1, 1) \cdot \mathbf{c}(\theta) = 0,$$

on which the full standard 3×3 SU(3)-type generator basis acts naturally.

Sixth, it remains viable under Dirac propagation:

$$(i\gamma^\mu \partial_\mu - m)\Psi_q \in \text{span}\{1, i, j, ij\}.$$

Seventh, propagation generates the mixed sector automatically:

$$\mathcal{S}_{ij} = \frac{1}{4} \gamma^\mu u A(\partial_\mu \theta) \cos\left(\frac{\theta}{4}\right).$$

Therefore the present construction yields a new internally viable SU(3) carrier wave for the quark sector. Its strong/color content is embedded directly into the phase structure of the wavefunction rather than appended as an external label; meson neutralization and baryon triplet closure emerge naturally from the carrier; the extracted color basis supports the full standard SU(3)-type matrix action; and the entire structure remains compatible with the Dirac shell through a closed $\{1, i, j, ij\}$ carrier algebra. This completes the construction of the new carrier and establishes it as a viable starting point for the next stage of the program.

4 Conceptual Position of the New Carrier

The preceding section established the new quark carrier, tested its closure properties, extracted the three phase classes, showed how meson and baryon formation arise from two distinct closure modes, recovered the standard 3×3 SU(3)-type generator basis from the extracted color space, and verified compatibility with Dirac propagation. The purpose of the present section is different. Rather than repeating those derivations, it clarifies what this construction changes conceptually, how it should be positioned relative to standard approaches, and why its route is structurally distinct.

4.1 What the present construction changes

The central change introduced by this paper is a change in the logical order of the quark-color story. In the standard presentation, one begins with an ordinary Dirac fermion and appends color as an internal SU(3) representation label. [3, 11, 8] In the present construction, the order is reversed. The carrier itself first generates a deeper phase structure from which three distinct phase classes and their opposite sectors are extracted. Only afterward does the standard color-matrix language appear as a representation acting on that extracted basis.

This difference is not cosmetic. In the present framework, color is not introduced as a primitive bookkeeping tag attached to an already complete carrier. Instead, it is interpreted as the visible readout of a deeper internal phase sector of the quark wavefunction. The recovered SU(3)-type matrix structure is therefore preserved, but its role changes. It is no longer the first source of the color sector. It becomes the matrix representation of a carrier structure that has already been established at the wavefunction level.

4.2 How this differs from the standard route

The usual quark-field route may be summarized schematically as

$$\text{Dirac carrier} + \text{color index} \longrightarrow \text{SU(3) representation structure.} [3, 11]$$

The present paper proceeds differently:

$$\text{extended carrier wave} \longrightarrow \text{extracted phase classes} \longrightarrow \text{SU(3)-type matrix representation.}$$

This difference is the defining feature of the construction. The standard route is label-first. The present route is carrier-first. The standard route begins from representation and assigns color internally to the quark field from the start. The present route begins from an extended phase carrier and asks whether the familiar color structure can be recovered as a consequence of that deeper wave object.

The distinction is especially important because the present carrier does more than reproduce a three-component internal space. It also produces two different closure modes from the same underlying wavefunction:

$$\text{mesons} \rightarrow \text{pair neutralization}, \quad \text{baryons} \rightarrow \text{triplet closure.}$$

Thus the carrier is not merely furnishing a new basis for the same old labels. It is supplying a common structural origin for the two basic hadronic existence modes.

4.3 Relation to other non-label intuitions

The present work does not belong to the usual label-first route, but neither does it discard the standard matrix formalism. Instead, it occupies an intermediate and more structural position. The familiar SU(3)-type generators are retained, but they are pulled out only after the phase carrier has already produced the color basis. [4] In this sense, the paper is not anti-representation. It is anti-primitive-label.

The broader intuition behind the construction is that a sufficiently rich wavefunction should not merely carry labels assigned from outside. It should begin to determine what sectors can exist, what sectors neutralize, what sectors close collectively, and what sectors survive as observable residues. That is the role the present carrier begins to play. Meson formation is interpreted as pairwise neutralization of the new sector, baryon formation as collective triplet closure of that same sector, and the standard Dirac-like part survives as the observable residue of composite closure.

4.4 Why Dirac is preserved rather than discarded

A second defining feature of the present construction is that Dirac is preserved rather than replaced. [1, 2] The original Dirac-like phase carrier

$$e^{i\theta}$$

remains present throughout. After meson neutralization or baryon closure, it is precisely this sector that survives as the observable residue. The new contribution

$$j \sin\left(\frac{\theta}{4}\right)$$

does not destroy the Dirac shell. It extends it into the quark/color sector.

This gives the construction a natural hierarchy:

$$\begin{aligned} 1, i &\Rightarrow \text{observable Dirac-type residue,} \\ j &\Rightarrow \text{new strong/color carrier sector,} \\ ij &\Rightarrow \text{mixed propagated sector.} \end{aligned}$$

The present paper should therefore be read neither as a rejection of Dirac nor as a rejection of the standard SU(3) matrix language. It is a proposal that both can be preserved while being placed in a deeper order: Dirac remains the observable shell, the new carrier supplies the quark/color phase structure, and the SU(3)-type matrices emerge as the representation of the extracted color basis.

4.5 Selection and existence

One of the most important consequences of this carrier-first route is that the wavefunction begins to act as a selector of what can and cannot survive as an observable hadronic structure. Pairwise neutralization supports meson-like existence. Collective triplet closure supports baryon-like existence. By contrast, non-closing or over-canceling configurations are naturally suggested to be unstable or unobservable.

The present paper does not claim to derive a full hadronic spectrum from this principle alone. However, it does establish the beginning of such a selection logic directly at the wavefunction level. In this sense, the carrier is no longer merely descriptive. It becomes partially constitutive of the logic of hadronic existence.

4.6 Connection to the broader hadronic program

The construction developed here addresses the quark-side problem: how a deeper strong/color sector may be embedded directly into the carrier wavefunction and how meson and baryon closure emerge from that embedding. It does not eliminate the need for configuration-level mass physics. Rather, it complements it.

The broader hadronic mass framework already addresses the question of how realized composite configurations are weighted energetically once a hadronic state is formed. [9] The present carrier supplies the missing quark-side origin for why distinct closure classes and composite residues exist at all. In that sense, the two frameworks divide labor naturally:

- the new carrier explains the deeper phase structure of quarks and the emergence of meson and baryon closure modes;
- the mass framework explains how the resulting realized composite configurations are interpreted at the hadronic level.

This division of roles is conceptually clean and preserves the strengths of both constructions.

4.7 Overall significance

The significance of the present construction is therefore not merely that it introduces a new formula. Its significance is that it shifts the quark-color problem from a label-first representation scheme to a carrier-first wavefunction scheme. The resulting structure preserves the standard SU(3)-type matrix language, preserves the observable role of Dirac, unifies meson and baryon formation under two closure modes of one carrier, and provides a deeper starting point for understanding how the strong/color sector may arise in wavefunction form.

5 Conclusion

This paper has constructed and tested a new SU(3) carrier wave for the quark sector by extending the ordinary Dirac phase carrier with one new perpendicular internal phase generator. The resulting quark carrier,

$$\Psi_q(x) = u(x) A(x) \left(e^{i\theta(x)} + j \sin\left(\frac{\theta(x)}{4}\right) \right),$$

together with the algebra

$$i^2 = j^2 = -1, \quad ij = ji, \quad (ij)^2 = +1,$$

was shown to satisfy the central structural demands placed on it in this work.

The carrier realizes the next closure rung through 8π -closure, generates three distinct phase classes directly from the new j -sector, produces anti-color by sign inversion, supports meson formation through pairwise neutralization, supports baryon formation through collective triplet closure, yields an extracted color basis on which the standard SU(3)-type 3×3 generators act naturally, and remains closed under Dirac propagation within the finite algebra

$$\text{span}\{1, i, j, ij\},$$

with the mixed ij -sector generated dynamically rather than inserted by hand.

These results support the main claim of the paper: the strong/color structure of the quark sector can be embedded directly into the phase content of the wavefunction itself rather than introduced only as an appended representation label. In the present construction, the standard SU(3) matrix picture is not discarded, but recovered at the correct stage. The matrices act on an extracted carrier basis rather than on a primitive color assignment. Mesons and baryons likewise do not enter as separate external organizational rules; they arise as two distinct closure modes of one common carrier.

Just as importantly, the construction preserves Dirac rather than replacing it. The original Dirac-like phase sector remains throughout and survives as the observable residue of composite closure. The new j -sector therefore acts not as a replacement for the Dirac shell, but as its extension into the quark/color domain. In this sense, the present proposal should be read as an attempt to complete the carrier structure of the quark sector while preserving the core success of the Dirac framework in the fermionic sector more broadly.

The present work is intentionally limited in scope. It does not claim to replace the full dynamical program of QCD, nor to derive an entire hadronic spectrum from first principles. What it does provide is a new internally viable quark carrier with an explicit wavefunction-level origin for color classes, anti-color opposition, meson neutralization, baryon triplet closure, extracted SU(3)-type representation, and Dirac-compatible propagation. That alone is a substantial result.

The broader implication is straightforward. If the quark sector does admit a deeper carrier origin than the standard label-first route suggests, then the familiar $SU(3)$ structure of the strong sector may be understood not merely as a representation appended to quarks after the fact, but as the matrix expression of a deeper phase-carrier geometry. The construction developed here establishes that possibility in explicit form and provides a concrete starting point for further development.

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